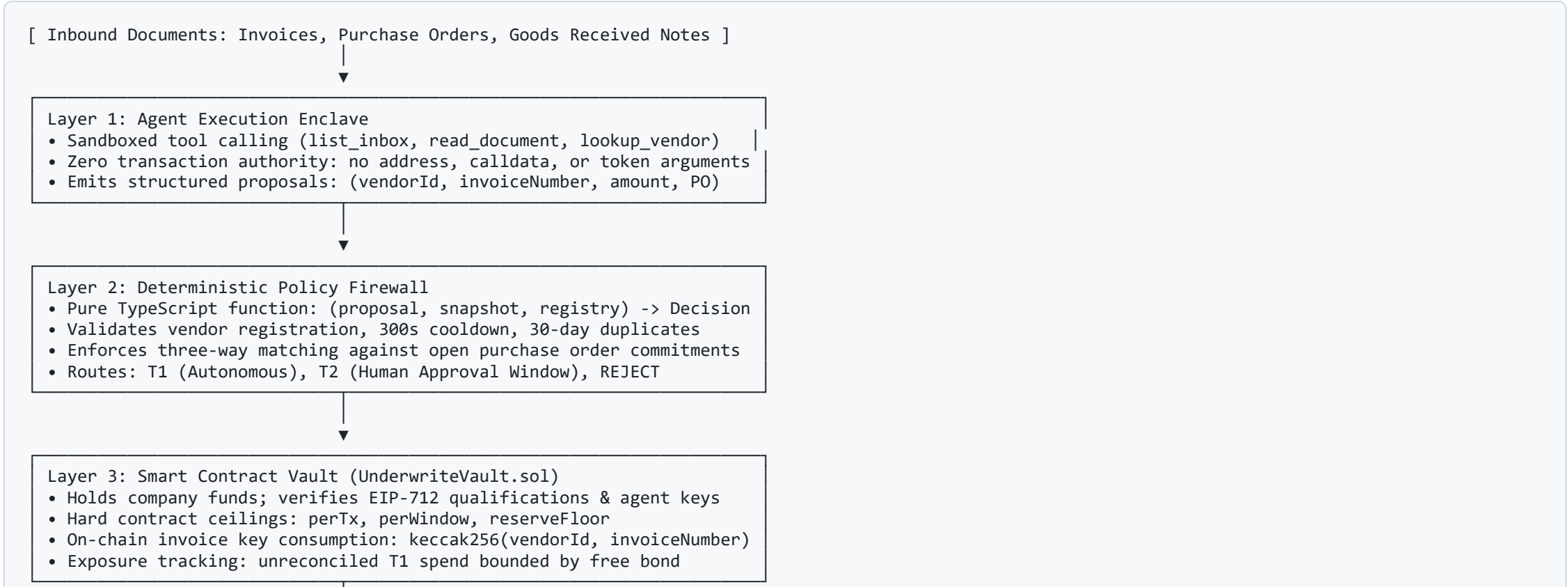
UNDERWRITE Technical Specification & Architecture

> **A Published, Verifiable Compiler from Adversarial Evidence to Collateralized Autonomous Authority**
>
> FinTechathon 2026 — Topic F Technical Reference

1. System Architecture

UNDERWRITE is an accounts-payable (AP) autonomous agent wallet designed for small businesses settling recurring supplier invoices in stablecoin. The system enforces strict separation of concerns across four non-overlapping security layers:



2. The Risk-to-Authority Compiler (`compiler@1`)

2.1 Core Problem & Principle

In standard agent wallets, spending limits are arbitrary numbers chosen by an operator without empirical justification. If an operator sets a $1,000 limit on a fragile agent, the business is exposed to $1,000 in catastrophic loss.

UNDERWRITE replaces operator guesswork with **evidence-sized authority**: the agent's autonomous spending cap is computed directly from its measured error rate on an adversarial examination.

2.2 Mathematical Formulation

The compiler operates as a pure, deterministic function mapping exam outcomes and owner risk preferences to an on-chain spending envelope:

$$\text{compiler@1}(\text{Report}, \text{OwnerInputs}) \longrightarrow (T_1^{\text{maxPerTx}}, T_1^{\text{maxPerWindow}}, T_2^{\text{allowed}}, \text{Status})$$

3. Landscape & Prior-Art Differentiation

3.1 Prior-Art Comparison Matrix

Solution	Evaluation Method	Financial Enforcement	Dynamic Qualification	Financial Recourse	Limitation	UI Ac
Coinbase Agentic Wallets (Feb 2026)	None public	Policy engine session caps	No (static config)	None	Caps are operator-chosen guesses	Ev de bc au
MetaMask Advanced Permissions (ERC-7715)	N/A	On-chain caveat enforcers	No	None	User manually grants arbitrary budgets	En th is ba

4. Prototype vs. Production Architecture

Dimension	Hackathon Prototype (Delivered)	Production Roadmap
Adjudication	Designated adjudicator key with hashed evidence	Decentralized oracle (e.g., UMA / Kleros) or multi-sig dispute board
Identity & Attestation	EIP-712 attestation with on-chain `examCommit` check	Confidential VM / TEE attestation (Intel SGX / AWS Nitro) bound to git commit
Underwriting Capital	Staked MockUSDC in `BondPool.sol` by exam author	Regulated institutional syndicate with dynamic yield staking
Chain Reach	Base Sepolia testnet + local Anvil enclaves	Multi-chain deployment (Arbitrum, Optimism, Base mainnet)

5. Formal System Limitations

1. **In-Distribution Scope:** Clopper-Pearson bounds apply strictly to the scenario distribution sampled by the exam generator. As documented in our calibration experiment (Hypothesis H4), statistical bounds on known threat families cannot mathematically guarantee defense against entirely novel, out-of-distribution adversarial vectors.
2. **Key Custody in Demo:** In the local demo runner, agent and owner private keys reside in local environment files (`.env`). Production deployments require hardware security modules (HSM) or smart contract account abstraction with WebAuthn/session keys.
3. **No Insurance Claims:** UNDERWRITE provides smart-contract-collateralized bond backing for autonomous exposure. It is **not** an insurance policy and makes no actuarial premium or insurance underwriting claims.